(i)
Pardeep twice, Quentin once:

$$P_{PPQ} = q\left(1-(1-p)^2\right) = q(2p-p^2)$$

Quentin twice, Pardeep once:

$$P_{PQQ} = p\left(1-(1-q)^2\right) = p(2q-q^2)$$

$$P_{PPQ} - P_{PQQ} = q(2p-p^2) - p(2q-q^2) = pq(q-p) > 0$$

$\therefore$ Rosalind should play Pardeep twice.

(ii)
Strategy 1:

$$P_1 = q\left(1-(1-p)^3\right) = pq(3-3p+p^2)$$

Strategy 2:

$$P_2 = \left(1-(1-p)^2\right)\left(1-(1-q)^2\right) = pq(2-p)(2-q)$$

Strategy 3:

$$P_3 = p\left(1-(1-q)^3\right) = pq(3-3q+q^2)$$

$$P_1 - P_3 = pq(3-3p+p^2) - pq(3-3q+q^2) = pq(q-p)(3-p-q) > 0$$

$$\begin{aligned} P_1 - P_2 &= pq(3-3p+p^2) - pq(2-p)(2-q) \\ &= pq\left(p-1+(q-p)(2-p)\right) \end{aligned}$$

For $q-p > \dfrac{1}{2}$,

$$p-1+(q-p)(2-p) > p-1+\frac{1}{2}(2-p) = \frac{p}{2}$$

$$\implies P_1 - P_2 > 0$$

$\therefore$ Rosalind should choose strategy 1.

For $q-p < \dfrac{1}{2}$, let $p = \dfrac{1}{2},\ q = \dfrac{3}{4}$

$$p-1+(q-p)(2-p) = \frac{1}{2} - 1 + \frac{1}{4}\cdot\frac{3}{2} = -\frac{1}{8} < 0$$

$$\implies P_1 < P_2$$

Strategy 2 is better than strategy 1

Let $p = \frac{1}{2},\ q = \frac{7}{8}$

$$p - 1 + (q - p)(2 - p) = \frac{1}{2} - 1 + \frac{3}{8} \cdot \frac{3}{2} = \frac{1}{16} > 0$$

$$\implies P_1 > P_2$$

Strategy 2 is worse than strategy 1

This solution was submitted by **Jon**

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